Pancreatitis (Acute and Chronic)

All key points for the exam + drug summary table

1. Overview: Acute vs Chronic Pancreatitis

Feature	Acute pancreatitis (AP)	Chronic pancreatitis (CP)
Definition	**Sudden** inflammation of the pancreas • usually **reversible**	**Progressive, irreversible** inflammation and **fibrosis** → loss of exocrine and endocrine function
Main causes	**Gallstones** (#1), **alcohol** (#2), **hypertriglyceridemia** (#3)	**Alcohol** (#1), **smoking**, genetic, recurrent acute pancreatitis
Pain	Sudden, severe, constant **epigastric pain radiating to the back**	Chronic or recurrent epigastric pain, worse after meals (may disappear late — "burnout")
Enzymes	**Lipase/amylase ≥ 3 × ULN**	Often **normal**
Key problems	Fluid loss, organ failure, necrosis, infection	**Steatorrhea**, weight loss, **diabetes**, chronic pain, fat-soluble vitamin deficiency
Main treatment	**IV fluids (Lactated Ringer's)**, **pain control**, **early feeding**, treat the cause	**Stop alcohol and smoking**, **pain management**, **pancreatic enzymes (PERT)**, vitamins, diabetes care

2. Acute Pancreatitis: Causes

Cause	Key points
Gallstones	**Most common cause** • stone blocks the ampulla • ↑ ALT (> 150 U/L suggests gallstones), ↑ bilirubin
Alcohol	**Second most common** • usually after years of heavy drinking
Hypertriglyceridemia	**TG > 1,000 mg/dL** (risk rises a lot above 1,000) • uncontrolled diabetes, obesity, pregnancy, drugs
Post-ERCP	Most common complication of ERCP
Drugs	See table below (~1–2% of cases)
Hypercalcemia	Hyperparathyroidism, malignancy
Other	Trauma, infections (mumps, CMV), **autoimmune (IgG4)**, genetic, pancreatic tumors, scorpion sting, smoking (risk factor), idiopathic

Mnemonic: I GET SMASHED — **I**diopathic, **G**allstones, **E**thanol, **T**rauma, **S**teroids, **M**umps, **A**utoimmune, **S**corpion sting, **H**ypercalcemia/**H**ypertriglyceridemia, **E**RCP, **D**rugs.

Drugs associated with acute pancreatitis (high-yield)

Class	Drugs
Immunosuppressants / IBD drugs	**Azathioprine**, **6-mercaptopurine**, **mesalamine**, **sulfasalazine**, corticosteroids
Antiepileptics	**Valproic acid** (boxed warning)
Antiretrovirals	**Didanosine** (boxed warning), stavudine, protease inhibitors (via ↑ TG)
Antimicrobials	**Pentamidine**, **sulfonamides (TMP-SMX)**, **tetracyclines**, **metronidazole**
Diuretics	**Furosemide**, **thiazides** (hydrochlorothiazide)
Diabetes drugs	**GLP-1 receptor agonists** (exenatide, liraglutide, semaglutide), **DPP-4 inhibitors** (sitagliptin) — association; **stop if pancreatitis is suspected**; avoid if history of pancreatitis
Drugs that ↑ triglycerides	**Estrogens** (oral contraceptives, HRT), **tamoxifen**, **isotretinoin**, **propofol** (lipid emulsion), protease inhibitors, atypical antipsychotics
Chemotherapy	**L-asparaginase / pegaspargase**, cytarabine
Others	**ACE inhibitors** (enalapril), **codeine**, simvastatin, octreotide (rare)

3. Acute Pancreatitis: Diagnosis and Severity

Diagnosis (Revised Atlanta) — 2 of 3

1. **Typical abdominal pain**: acute, severe, persistent **epigastric pain**, often **radiating to the back**, with nausea/vomiting.
2. **Serum lipase or amylase ≥ 3 × upper limit of normal** (**lipase preferred**: more specific and stays elevated longer).
3. **Imaging findings** (contrast CT, MRI, or ultrasound).

- **Enzyme level does not correlate with severity** and is **not used to monitor** or decide feeding.
- **CT is not needed** when the diagnosis is clear; do contrast CT if the diagnosis is unclear or **no improvement after 48–72 hours** (to look for necrosis).
- **Abdominal ultrasound** for **all patients** to look for **gallstones**.
- Check **triglycerides** and **calcium** if no gallstones and no alcohol history.
- Signs of severe hemorrhagic pancreatitis: **Cullen sign** (periumbilical bruising), **Grey Turner sign** (flank bruising).

Severity (Revised Atlanta)

Severity	Definition	Mortality
Mild (most common, ~80%)	**No organ failure**, no local or systemic complications	Very low
Moderately severe	**Transient organ failure (< 48 h)** and/or local complications or worsening of comorbidity	Low
Severe	**Persistent organ failure (> 48 h)** — respiratory, cardiovascular, or renal	High (up to 30%)

Predicting severity

- **SIRS** at admission and persisting > 48 h → higher mortality.
- **BISAP score** (≥ 3 = high risk): **B**UN > 25 mg/dL, **I**mpaired mental status, **S**IRS, **A**ge > 60, **P**leural effusion.
- Other markers: **rising BUN** or **hematocrit > 44%** (hemoconcentration), **CRP > 150 mg/L at 48 h**, APACHE II ≥ 8, Ranson criteria (older), obesity, older age.

Local complications

Complication	Timing / description
Acute peripancreatic fluid collection	< 4 weeks, no wall • usually resolves
Pancreatic pseudocyst	**> 4 weeks**, well-defined wall, fluid only • **asymptomatic → observe** • symptomatic → endoscopic drainage
Acute necrotic collection	< 4 weeks, contains necrosis
Walled-off necrosis (WON)	**> 4 weeks**, encapsulated necrosis
Infected necrosis	Usually after **7–10 days** • fever, worsening, **gas on CT** → antibiotics ± drainage
Others	Splenic vein thrombosis, pseudoaneurysm, abdominal compartment syndrome

Systemic complications

ARDS, **AKI**, **shock**, **hypocalcemia** (calcium binds fatty acids — **fat saponification**; check Chvostek/Trousseau signs), **hyperglycemia**, DIC, worsening of comorbidities.

4. Acute Pancreatitis: Management

A) Fluid resuscitation (most important early step)

- **Lactated Ringer's (LR) is preferred** over normal saline (less SIRS; NS can cause hyperchloremic acidosis) • avoid LR in **hypercalcemia**.
- **Moderate, goal-directed** fluids (ACG 2024, **WATERFALL trial**): **10 mL/kg bolus** only if hypovolemic, then **1.5 mL/kg/h** — **aggressive fluids** (20 mL/kg bolus + 3 mL/kg/h) caused **more fluid overload** without benefit.
- Reassess frequently (6 h, 12–24 h, 48 h) • **targets:** HR < 120, MAP 65–85 mmHg, **urine output > 0.5–1 mL/kg/h**, **falling BUN and hematocrit**.
- Careful in **heart failure**, **CKD**, elderly (watch for pulmonary edema, abdominal compartment syndrome).

B) Pain control

- **Opioids are safe and effective**: **hydromorphone**, **morphine**, **fentanyl** (IV, often as **PCA**) • the old concern that morphine worsens pancreatitis (sphincter of Oddi spasm) is **not clinically important**.
- **Avoid meperidine** (normeperidine → **seizures**, especially in renal failure).
- **Multimodal: acetaminophen** and **NSAIDs** (if no AKI) reduce opioid needs • epidural analgesia in severe cases.

C) Nutrition

- **Mild AP: early oral feeding within 24–48 hours** as tolerated (pain decreasing) with a **low-fat solid diet** — no need to start with clear liquids or wait for lipase to normalize • early feeding ↓ hospital stay.
- **Severe AP: enteral nutrition** (nasogastric or nasojejunal — both acceptable) is **preferred over parenteral (TPN)** → protects the gut barrier, ↓ infection, organ failure, and death.
- **TPN** only if enteral feeding is not tolerated or not possible.

D) Antibiotics

- **NO prophylactic antibiotics** — not in severe AP and **not in sterile necrosis**.
- **Give antibiotics only for infection: infected necrosis** (suspect if fever/clinical worsening after 7–10 days, gas on CT) or extrapancreatic infection (**cholangitis**, pneumonia, UTI, bacteremia).
- **Choice (penetrate necrotic tissue): carbapenems** (**meropenem**, imipenem), **fluoroquinolones + metronidazole**, or piperacillin-tazobactam.
- **Infected necrosis:** antibiotics first, then **delayed** (ideally **> 4 weeks**, after walling-off) **minimally invasive drainage/necrosectomy** ("step-up approach").

E) Treat the cause

Cause	Management
Gallstone pancreatitis	**Urgent ERCP (within 24 h)** only if **cholangitis** (fever, jaundice, RUQ pain) or persistent bile duct obstruction • Not needed in mild AP without obstruction • **Cholecystectomy during the same admission** (mild AP) to prevent recurrence • necrotizing AP: delay until resolved
Alcohol	**Brief alcohol counseling** during admission • monitor for **alcohol withdrawal** (CIWA; benzodiazepines) • **thiamine** before glucose • folate, multivitamin
Hypertriglyceridemia (TG > 1,000)	**Insulin infusion** (± dextrose to prevent hypoglycemia) → activates **lipoprotein lipase** → ↓ TG (especially with hyperglycemia) • **Plasmapheresis** in severe cases • Heparin (controversial) • Long term: **fibrates** (fenofibrate, gemfibrozil), **omega-3**, **low-fat diet**, diabetes control, **no alcohol**, stop TG-raising drugs • goal **TG < 500**
Hypercalcemia	Treat cause (e.g., parathyroidectomy)
Drug-induced	**Stop the drug** and do not rechallenge

F) What does NOT work (not recommended)

- **Prophylactic antibiotics** • **octreotide/somatostatin** • protease inhibitors (gabexate) • **probiotics** (harmful in severe AP) • routine **PPIs** • **keeping the patient NPO** until lipase normalizes.

G) Supportive care

- **VTE prophylaxis** (heparin/enoxaparin) • glucose control • oxygen • electrolyte replacement (**calcium**, magnesium, potassium) • ICU for organ failure.

H) Preventing post-ERCP pancreatitis (high-yield)

- **Rectal indomethacin 100 mg** (or rectal diclofenac 100 mg) **immediately before or after ERCP**.
- **Aggressive Lactated Ringer's hydration** around the procedure.
- **Pancreatic duct stent** in high-risk patients.

5. Chronic Pancreatitis

Causes (TIGAR-O)

- **Toxic-metabolic: alcohol** (most common), **smoking** (independent risk; synergistic with alcohol), hypercalcemia, hypertriglyceridemia, CKD.
- **Idiopathic** • **Genetic** (PRSS1, SPINK1, **CFTR** — cystic fibrosis) • **Autoimmune** (IgG4) • **Recurrent acute pancreatitis** • **Obstructive** (strictures, tumors, pancreas divisum).

Clinical features

Feature	Details
Pain	Recurrent/chronic epigastric pain radiating to the back, **worse after meals** • may decrease late in disease
Exocrine insufficiency	**Steatorrhea** (oily, foul-smelling, floating stools; appears when lipase output < 10%), bloating, **weight loss**, malnutrition, **fat-soluble vitamin (A, D, E, K) deficiency**, **osteoporosis**
Endocrine insufficiency	**Diabetes (type 3c, pancreatogenic)** — loss of insulin **and glucagon** → **brittle** glucose control, ↑ **hypoglycemia risk**
Classic triad (late)	**Pancreatic calcifications** + **steatorrhea** + **diabetes**

Diagnosis

- **Imaging:** CT (**calcifications**, ductal dilation, atrophy) • MRI/MRCP (± secretin) • endoscopic ultrasound (EUS).
- **Fecal elastase-1 < 200 mcg/g** → exocrine insufficiency (< 100 = severe) • 72-hour fecal fat (rarely done).

- **Amylase/lipase are often normal** (burned-out pancreas).
- Screen for **diabetes** (HbA1c, fasting glucose) yearly, **vitamin D** and fat-soluble vitamins, **bone density**.

Complications

Pseudocyst, **bile duct or duodenal obstruction**, **splenic vein thrombosis** (→ isolated gastric varices), **pancreatic cancer** (↑ risk, especially hereditary), malnutrition, **osteoporosis**, **opioid dependence**.

6. Chronic Pancreatitis: Management

A) Lifestyle (most important)

- **Stop alcohol completely** and **stop smoking** → slows progression and may ↓ pain.
- **Diet:** small frequent meals • **normal-fat diet with enzymes** is preferred (very low-fat diets worsen malnutrition) • medium-chain triglycerides (MCT oil) if needed • dietitian referral.

B) Pain management (stepwise)

Step	Options
1. Non-opioid analgesics	**Acetaminophen** • **NSAIDs** (short-term, watch GI/renal)
2. Neuropathic adjuvants	**Pregabalin** (**best evidence**) • gabapentin • **TCAs** (amitriptyline, nortriptyline) • **SNRIs** (duloxetine) • SSRIs for comorbid depression
3. Weak opioid	**Tramadol** (preferred over morphine: similar pain relief, fewer GI side effects) • risk of **seizures** and **serotonin syndrome** (with SSRIs/SNRIs)
4. Strong opioids	Last resort, lowest dose, opioid agreement • risk of dependence and **narcotic bowel**
Other	**Antioxidants** (selenium, vitamins C and E, methionine) — modest benefit • **PERT** is not recommended for pain alone • **Endoscopic therapy** (stones, strictures; ESWL) • **celiac plexus block** • **surgery** (Puestow, Frey, Whipple, total pancreatectomy with islet autotransplant)

C) Pancreatic enzyme replacement therapy (PERT)

- **Indication: exocrine pancreatic insufficiency** — steatorrhea, weight loss, malnutrition, low fecal elastase (also cystic fibrosis, pancreatic cancer, pancreatectomy).
- **Product: pancrelipase** (lipase + protease + amylase from **pig (porcine) pancreas**) • **dosed by LIPASE units**.
- **Brands: Creon**, **Zenpep**, **Pancreaze**, **Pertzye** (enteric-coated capsules) • **Viokace** (**non-enteric-coated tablet → must be given with a PPI**; adults only) • brands are **not interchangeable**.
- **Relizorb**: lipase cartridge for **enteral (tube) feeding** (microbial lipase, not porcine).

Population	Lipase dose
Adults (chronic pancreatitis)	**40,000–50,000 USP lipase units with each meal** and **half (20,000–25,000) with each snack** • titrate up (e.g., to 75,000–90,000 per meal) based on response
Weight-based (all ages)	Start **500 units/kg/meal** (≥ 4 years; 1,000 units/kg/meal if < 4 years) • **Max 2,500 units/kg/meal** (or **10,000 units/kg/day**, or 4,000 units/g fat/day)
Infants	**2,000–4,000 units per 120 mL** of formula or breastfeeding

Administration and counseling (high-yield)

- Take **with the first bite of every meal and snack** (or half at the start and half during a long meal) — **not** before or after.
- **Swallow whole** — **do not crush or chew** the capsule or its beads.
- If unable to swallow: **open the capsule** and sprinkle on a small amount of **acidic soft food** (pH < 4.5, e.g., **applesauce**); swallow **immediately without chewing**, then drink water • **do not mix with milk/dairy** or alkaline foods (dissolves the coating).

- **Do not hold in the mouth** (oral mucosal irritation).
- Not needed with drinks or snacks that contain no fat (e.g., fruit, juice).
- Store at room temperature, keep dry.

Poor response to PERT — check

1. **Adherence** and **timing** with meals
2. **Dose** too low → increase
3. **Stomach acid** inactivating enzymes → add a **PPI or H2RA** (enteric coating dissolves at pH > 5.5; ↓ duodenal bicarbonate in CP)
4. **Other causes:** small intestinal bacterial overgrowth (SIBO), celiac disease, bile acid diarrhea

Adverse effects and interactions

- Abdominal pain, nausea, flatulence, diarrhea or constipation, headache.
- **Fibrosing colonopathy** (colonic strictures) with **high doses** (> 6,000 units/kg/meal), mainly in children with cystic fibrosis → **do not exceed 2,500 units/kg/meal**.
- **Hyperuricemia**/gout (purines) at high doses • **pork allergy** • theoretical viral transmission (porcine source).
- **Religious/cultural:** products are **porcine** → discuss with Muslim and Jewish patients (generally permitted when no alternative exists; Relizorb is non-porcine for tube feeds).
- **DDIs:** ↓ effect of **acarbose** and **miglitol** • calcium/magnesium antacids may ↓ effect of non-enteric-coated products • **PPIs/H2RAs** improve efficacy.

Monitoring

Symptoms (**stool frequency and consistency**, steatorrhea, bloating), **weight**, nutritional status (**vitamins A, D, E, K**, prealbumin, zinc), bone density.

D) Nutritional supplements

- **Fat-soluble vitamins (A, D, E, K)** as needed • **calcium + vitamin D**.
- **Bone density (DEXA)** every ~2 years (high osteoporosis risk).
- Check vitamin B12, zinc, magnesium.

E) Diabetes (type 3c)

- **Metformin** first-line if no contraindication (may ↓ pancreatic cancer risk).
- **Insulin** often needed • **higher hypoglycemia risk** (glucagon deficiency, malnutrition, alcohol) → careful titration and education.
- **GLP-1 RAs and DPP-4 inhibitors** — generally **avoided** (pancreatitis concerns); SGLT2 inhibitors — caution (ketoacidosis, volume depletion).
- Take PERT consistently to avoid fluctuating glucose absorption.

F) Autoimmune pancreatitis (IgG4-related)

- Elevated **IgG4**, "sausage-shaped" pancreas, may mimic pancreatic cancer.
- **Responds to corticosteroids: prednisone 0.6–1 mg/kg/day (≈ 40 mg/day) for ~4 weeks**, then taper • **rituximab** for relapse or steroid-dependence.

Exam Traps and Frequently Tested Points

1. **Acute pancreatitis diagnosis: 2 of 3** — typical pain, **lipase/amylase ≥ 3 × ULN**, imaging.
2. **Lipase** is more specific than amylase; enzyme levels **do not reflect severity**.
3. **Most common causes: gallstones** > alcohol > hypertriglyceridemia (**TG > 1,000**).
4. **Ultrasound** in all patients (gallstones) • CT only if unclear or no improvement in 48–72 h.
5. **Drug causes: azathioprine/6-MP**, **valproic acid**, **didanosine**, **pentamidine**, **TMP-SMX**, **tetracyclines**, **furosemide/thiazides**, **mesalamine/sulfasalazine**, **estrogens**, **L-asparaginase**, **GLP-1 RAs/DPP-4 inhibitors**, ACE inhibitors, propofol.
6. **Severity:** mild (no organ failure), moderately severe (organ failure < 48 h), **severe (persistent organ failure > 48 h)**.
7. **BISAP ≥ 3**, persistent SIRS, rising BUN, Hct > 44% → severe disease.

8. **Fluids: Lactated Ringer's preferred** • **moderate** (10 mL/kg bolus if hypovolemic + 1.5 mL/kg/h), not aggressive.
9. **Pain:** opioids (hydromorphone, morphine, fentanyl) are fine • **avoid meperidine**.
10. **Mild AP: early oral feeding (24–48 h)**, low-fat solid diet — don't wait for lipase.
11. **Severe AP: enteral > parenteral nutrition** (NG or NJ).
12. **No prophylactic antibiotics** (even severe or sterile necrosis) • treat only **infected necrosis** or other infections → **carbapenem** or **quinolone + metronidazole**.
13. **Infected necrosis:** antibiotics + **delayed (> 4 weeks) minimally invasive drainage**.
14. **Gallstone AP: ERCP within 24 h only with cholangitis • cholecystectomy same admission** (mild).
15. **Hypertriglyceridemia AP: insulin infusion** (+ dextrose), plasmapheresis • then fibrates, omega-3, low-fat diet.
16. **Post-ERCP pancreatitis prevention: rectal indomethacin 100 mg** + LR hydration + pancreatic stent.
17. **Not recommended:** octreotide, probiotics, prophylactic antibiotics, prolonged NPO.
18. **Hypocalcemia** in AP from fat saponification; Cullen and Grey Turner signs = hemorrhagic pancreatitis.
19. **Pseudocyst** = > 4 weeks, fluid only • asymptomatic → observe.
20. **Chronic pancreatitis: alcohol** (#1) + **smoking** • triad: **calcifications, steatorrhea, diabetes**.
21. **Fecal elastase < 200 mcg/g** → exocrine insufficiency • amylase/lipase often normal in CP.
22. **Type 3c diabetes** → loss of glucagon → **hypoglycemia risk** • metformin first, insulin often needed.
23. **CP pain:** acetaminophen/NSAIDs → **pregabalin** (best evidence), TCAs, SNRIs → **tramadol** → strong opioids last.
24. **PERT (pancrelipase)** dosed by **lipase units**: adults **40,000–50,000 units/meal**, **half with snacks**.
25. **Max 2,500 lipase units/kg/meal** (10,000 units/kg/day) → **fibrosing colonopathy** at high doses.
26. **Take PERT with the first bite** of each meal/snack • swallow whole • can sprinkle on **applesauce** (not dairy) • don't hold in mouth.
27. **Poor PERT response** → check adherence/timing, ↑ dose, **add PPI**, rule out SIBO.
28. **Viokace** (non-enteric-coated) **requires a PPI** • PERT ↓ effect of **acarbose/miglitol**.
29. **PERT is porcine** (religious considerations).
30. **Supplement vitamins A, D, E, K**; check **bone density**.
31. **Autoimmune (IgG4) pancreatitis → corticosteroids**.

Note: Fluid rates, PERT doses, and recommendations differ slightly between references (ACG 2024, AGA). **Use the numbers in the reference you are studying for the exam.**

Drug Summary Table: Doses, Key ADRs and DDIs

Drug	Dose	Key ADRs	Key DDIs / special notes
Lactated Ringer's	AP: 10 mL/kg bolus if hypovolemic, then 1.5 mL/kg/h (goal-directed)	**Fluid overload**, pulmonary edema	Preferred over NS • avoid in **hypercalcemia** • contains calcium (do not co-infuse with ceftriaxone in neonates)
Hydromorphone	0.2–1 mg IV q2–3h PRN or PCA	Sedation, respiratory depression, constipation, nausea	↑ CNS depression with benzodiazepines, alcohol • preferred in renal impairment over morphine
Morphine	2–4 mg IV q2–4h PRN or PCA	Same + histamine release	Active metabolites accumulate in renal failure
Fentanyl	25–100 mcg IV q1–2h PRN or infusion/PCA	Respiratory depression, chest wall rigidity (rapid IV)	**CYP3A4 substrate** (inhibitors ↑ levels) • safe in renal failure
Meperidine	Avoid	**Normeperidine → seizures**	**Serotonin syndrome with MAOIs** • not recommended

Drug	Dose	Key ADRs	Key DDIs / special notes
Acetaminophen	650–1,000 mg q6h (max 3–4 g/day; ≤ 2 g/day with heavy alcohol use/liver disease)	Hepatotoxicity (overdose)	↑ INR with warfarin (regular use)
NSAIDs (e.g., ibuprofen, ketorolac)	Ibuprofen 400–800 mg q6–8h • ketorolac 15–30 mg IV q6h (≤ 5 days)	**AKI**, GI bleeding	Avoid in AKI, GI bleed • ↑ bleeding with anticoagulants
Indomethacin (rectal)	**100 mg PR once** immediately before/after ERCP	GI bleeding (rare with single dose)	Prevents post-ERCP pancreatitis
Meropenem	1 g IV q8h (infected necrosis)	Diarrhea, C. diff, seizures (less than imipenem)	↓ **valproic acid levels** markedly (avoid combination)
Imipenem-cilastatin	500 mg IV q6h	**Seizures** (renal impairment), C. diff	↓ **valproic acid** • renal dose adjustment
Ciprofloxacin + metronidazole	Cipro 400 mg IV q12h (500–750 mg PO BID) + metronidazole 500 mg q8–12h	Tendinopathy, QT • metallic taste, neuropathy	Chelation (cipro) • **no alcohol** (metronidazole) • ↑ warfarin INR
Regular insulin infusion (HTG-pancreatitis)	0.1 units/kg/h IV (+ dextrose if glucose < 200)	**Hypoglycemia**, **hypokalemia**	Monitor glucose hourly, K • activates lipoprotein lipase
Fenofibrate	48–145 mg daily (product-dependent)	↑ SCr, ↑ LFTs, myopathy, gallstones	↑ myopathy with statins (less than gemfibrozil) • ↑ warfarin effect • renal adjustment
Gemfibrozil	600 mg BID before meals	GI upset, gallstones, myopathy	**Avoid with statins** (rhabdomyolysis) • **CYP2C8 inhibitor** (↑ repaglinide — CI) • ↑ warfarin
Omega-3 fatty acids (icosapent ethyl, omega-3-acid ethyl esters)	4 g/day	Fishy taste, GI upset, ↑ bleeding risk, atrial fibrillation	Caution with anticoagulants
Pancrelipase (Creon, Zenpep, Pancreaze, Pertzye)	Adults: 40,000–50,000 lipase units/meal, half with snacks • max 2,500 units/kg/meal (10,000 units/kg/day)	Abdominal pain, flatulence, **fibrosing colonopathy** (high dose), hyperuricemia	Take with first bite • swallow whole • ↓ **acarbose/miglitol** • PPI improves effect • porcine
Pancrelipase (Viokace)	Non-enteric-coated tablet with meals	Same + oral mucositis	**Must be given with a PPI** • adults only
Omeprazole / PPIs (with PERT)	20–40 mg daily	Headache, diarrhea • long-term: C. diff, ↓ Mg, ↓ B12, fractures	Improves PERT efficacy • CYP2C19 (omeprazole: ↓ clopidogrel)
Pregabalin	75 mg BID → up to 300 mg BID	**Dizziness**, **sedation**, **edema**, weight gain	Renal adjustment • ↑ CNS depression with opioids/alcohol • Schedule V
Gabapentin	300 mg HS → up to 1,200 mg TID	Dizziness, sedation	Renal adjustment • ↑ CNS depression with opioids
Amitriptyline / nortriptyline	10–25 mg HS, titrate (max ~75–100 mg)	**Anticholinergic**, sedation, orthostasis, **QT**	↑ serotonin syndrome with tramadol/SSRIs • avoid in elderly (Beers)
Duloxetine	30 mg daily → 60 mg daily	Nausea, dry mouth, hepatotoxicity	Avoid in **heavy alcohol use**/liver disease • serotonin syndrome with tramadol • CYP1A2 (fluvoxamine, cipro ↑)
Tramadol	50–100 mg q4–6h (max 400 mg/day; 200 mg if CrCl < 30)	**Seizures**, **serotonin syndrome**, nausea, hypoglycemia	Avoid with **SSRIs/SNRIs/TCAs/MAOIs** • CYP2D6 (inhibitors ↓ effect)

Drug	Dose	Key ADRs	Key DDIs / special notes
Vitamins A, D, E, K	Per deficiency (e.g., vitamin D3 1,000–2,000 IU/day or higher if deficient)	Toxicity with excess A and D	Vitamin K affects warfarin • vitamin E ↑ bleeding at high doses
Prednisone (autoimmune pancreatitis)	0.6–1 mg/kg/day (~40 mg) × 4 weeks, then taper	Hyperglycemia (worse in type 3c DM), osteoporosis, infection	Calcium + vitamin D • monitor glucose
Thiamine (alcohol use)	100 mg IV/PO daily (higher if Wernicke's suspected)	Rare (anaphylaxis IV)	Give **before glucose**
Lorazepam / chlordiazepoxide (alcohol withdrawal)	Symptom-triggered by CIWA score	Sedation, respiratory depression	Lorazepam preferred in liver disease • ↑ CNS depression with opioids

Good luck on your exam!